DECODING THE GRID: AN EXPLORATORY ANALYSIS OF TRAFFIC AND ACCIDENT PATTERNS IN THE CITY OF CHICAGO

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1. ABSTRACT

Abstracts should be about 200 words. All running text, including the abstract, should be right-justified, in two columns, single-spaced, and in Times New Roman size 10 font.

Keywords: Place any keywords here

2. INTRODUCTION

2.1 Background

Since the inception and production of the mass-affordable Model-T Ford in 1908, consumer cars have become a key figure of worldwide transportation- particularly in the United States, where they dominate as the primary method of personal commuting. Unfortunately, this reliance on automobiles comes with a hefty price of increased accident rates and consequent injuries. Motor vehicle crashes represent one of the most critical public health challenges facing the United States. In 2023, 40,901 people died in motor vehicle crashes, while millions more sustained injuries ranging from minor to permanently debilitating (IIHS 2025). These fatalities mark motor vehicle crashes as the one of the largest causes of accidental deaths nationally, In addition to the health concerns, there is also a large economic impact; the National Highway Traffic Safety Administration estimates that motor vehicle crashes cost Americans $340 billion annually, accounting for medical expenses, lost productivity, legal costs, emergency services, insurance administration, congestion impacts, and property damage (NHTSA 2023). When quality-of-life valuations are included in this calculation, the total societal harm approaches a staggering $1.4 trillion (NHSTA 2023).

Despite years of progress — through improved vehicle safety features, strict enforcement of seat belt laws, and enhanced roadway design — the traffic safety landscape has had some setbacks in recent years. After three decades of decline, roadway fatalities reversed course with 2021 recording 43,230 deaths, which is the highest number since 2005 (USDoT 2024). While recent data shows some improvements, the rate remains significantly higher than pre-pandemic levels and elevated compared to other nations (IIHS 2025). Contributing factors to this crisis include speeding, impaired driving, distracted driving, and failure to use safety restraints. Speeding alone was a factor in 29% of motor vehicle crash deaths in 2023, emphasizing the critical, even vital, need for effective enforcement and mitigation strategies (IIHS 2025).

In response to these challenges, many municipalities have implemented automated traffic enforcement systems, including speed cameras and red-light cameras, to supplement traditional policing efforts. However, the effectiveness of these interventions remains a subject of active research and debate. Studies have found that red light cameras reduce fatal red light running crash rates in large cities by 21%, though some research indicates they may increase rear-end collisions while decreasing more severe right-angle crashes (Hu 2025). Overall, the National Highway Traffic Safety Administration assigns automated traffic enforcement (ATE) the highest rating for effectiveness among all traffic safety countermeasures (Peterman 2020). However, mixed findings across different jurisdictions mean that an understanding of local elements is required to fully comprehend the impacts of these interventions.

Recent advances in machine learning and data analytics have created new methods for understanding and predicting traffic crash patterns, something explored further in this paper. Research has shown that supervised learning algorithms can effectively predict crash injury severity based on various contributing factors. Studies utilizing Random Forest and Logistic Regression models have achieved prediction accuracy of up to 87% when analyzing factors such as vehicle characteristics, driver demographics, temporal patterns, and road conditions (Obasi and Benson 2023). A comprehensive review spanning 56 studies also found that Random Forest was the most consistently effective algorithm, achieving optimal performance in 70% of applications, with Support Vector Machines, Decision Trees, and K-Nearest Neighbors also showing promise in a variety of applications (Sandos et. al. 2022). These predictive models enable transportation agencies to identify high-risk scenarios, allocate emergency response resources more effectively, and design targeted interventions to reduce crash severity.

The relationship between socioeconomic factors and traffic safety outcomes has also garnered increased attention in transportation research. Emerging evidence suggests that crash frequency, severity, and injury rates can vary systematically across neighborhoods with different poverty levels, though the underlying causes of these disparities remain incompletely understood. Uncovering and developing potential explanations for these relationships is essential for developing fair and useful transportation policies that address disparities in the conditions of different urban communities.

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2.2 Overview and Approach

This study presents an encompassing, comprehensive analysis of traffic accident data from the City of Chicago. By applying multiple data sources and analytical approaches, it aims to create a useful case study and advance understanding of urban traffic safety. The analysis encompasses four complementary research objectives designed to provide actionable insights into traffic safety planning.

First, the study evaluates the effectiveness of Chicago's automated enforcement infrastructure by examining the impact of speed cameras and red-light cameras on accident frequency and injury severity. This analysis breaks it down to the individual camera level, controlling for spatial characteristics and traffic volume to isolate the causal effects of camera installations on safety outcomes in their immediate vicinity.

Second, machine learning models are developed to predict accident injury severity based on a comprehensive set of environmental, temporal, spatial, and roadway characteristics. Multiple algorithms are trained and compared, including Random Forest, Logistic Regression, and Gradient Boosting methods, with model performance evaluated using standard classification metrics. Feature importance analysis provides the most influential potential predictors of severe injuries, giving insight into trends and a basis for targeted prevention strategies.

Third, the study investigates the relationship between neighborhood poverty rates and traffic safety outcomes, which were observed during exploratory data analysis. By joining crash records with Census data at the zip code level, the analysis examines whether areas with higher poverty concentrations experience elevated crash rates, more severe injuries, or both. Potential mediating factors— including road quality, average speed limit, and lighting—are explored to understand the causal pathways between income and adverse accident outcomes.

Fourth, an interactive geospatial visualization is developed to identify and display accident hotspots dynamically. Using unsupervised clustering algorithms, the system identifies locations with elevated crash concentrations, with the number of clusters adjusted dynamically based on user-specified parameters. The visualization then distinguishes between weekday and weekend patterns and allows users to examine temporal variations by hour of day, enabling transportation planners to identify when and where safety interventions are needed most.

Together, these analyses provide a holistic view of Chicago's traffic safety landscape, combining evaluation of existing interventions, predictive modeling capabilities, investigation of equity concerns, and practical decision-support tools. The findings have direct relevance, especially to the city of Chicago, for municipal transportation departments, law enforcement agencies, public health officials, and urban planners working to reduce the overall toll of traffic crashes in metropolitan areas.

2.3 Data Source

The primary data source consists of accident records and camera information directly from the City of Chicago (abbv. CoC) open data portal. Secondary socioeconomic and demographic data was also pulled from the U.S Census Bureau. In total, there were 6 tables acquired that were used in the final analysis:

- CoC Traffic Crash Records (1 table)
    - 911K rows, 48 columns, Sep 2017 - Oct 2025
    - Contains information from each crash on city streets within City of Chicago limits and under the jurisdiction of Chicago Police Department (CPD)
- CoC Red Light and Speeding Camera Locations (2 tables)
    - 360 total rows, 8 columns, 2004 - 2025
    - Contains information on each red light and speeding camera in the City of Chicago, including coordinates and go-live date
- CoC Red Light and Speeding Camera Violations (2 tables)
    - 1.5M total rows, 9 columns, 2014 – 2025
    - Contains information on violations captured on each red light and speeding camera
- US Census Bureau 5-Year American Community Survey
    - 61 rows, 6 columns, 2023
    - Contains select relevant socioeconomic variables from census data on the most recent 5-year average of the ACS within City of Chicago zip codes

Together, these tables formed the basis of data that would be used for all 4 elements of analysis.

3 METHODS AND MATERIALS

3.1 Data Cleaning and Processing

To prepare the data for analysis, data cleaning and preprocessing using Python and the Pandas library was conducted, with the preprocessing mainly entailing column renaming and creation, typecasting and categorical conversion to binary/one-hots, excluding nulls and/or imputing missing values, and dropping irrelevant columns. Spatial analysis was then applied to implement zip codes for each coordinate data

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point, which was used to join the Chicago accident and camera data with their respective zip codes.

3.1.1 Census and Demographic Data Preprocessing

The first part of the preprocessing focused on the census data, where the structure of the data was reformatted to align with the traffic data. To do this, shapefiles representative of zip code boundaries were incorporated to accurately map demographic statistics to specific geographic regions. The original dataset from the census featured obscure census variable codes, which were then renamed into human-readable columns, standardizing the formatting of zip code identifiers. This allowed data transformations to ensure that socioeconomic factors could be accurately mapped against the locations of crashes established in the crashes dataset.

3.1.2 Crash and Vehicle Data Preprocessing

To obtain usable crash, vehicle, and people data associated with traffic crashes in the City of Chicago, all the timestamp columns had to be converted into the proper datetime format, allowing a time-series analysis to be conducted on the data. Some of the data rows contained missing or invalid data. To address this for each dataset, the number of missing values for each section was analyzed and, in accordance with the analysis requirements for the project, entries lacking critical data points were flagged. Once all entries not containing information were flagged, it was determined that removing these columns would not affect the overall results, as the total amount of data available still far exceeded the values removed. To ensure the use of the same column names across the project data, the column names were standardized to ensure consistency across datasets.

3.1.3 Red Light and Speed Camera Preprocessing

The raw data for both the red light and speed cameras were initially presented as two separate datasets. One for the camera locations, and a separate dataset for violations. During the initial exploratory analysis, it was observed that the speed camera location dataset did not have any reference to the ***cameraID*** for each violation, rather each entry only included the cross-streets in which the violations were recorded. To merge the camera location data with its respective violations data, the violations dataset had to be manually updated to map respective intersections with those recorded in the camera locations dataset.

Once the proper mappings were obtained, a standard join function using the available identifiers was applied to merge the red light and speeding camera locations with their respective violation logs, creating two master files**,** ***df_redlights*** and ***df_speeding***, which contained both the historical violation volume, as well as precise geospatial coordinates for each camera. These two datasets were then unified to create ***df_cameras***, which became the dataset used for analysis.

3.2 Exploratory Data Analysis

3.2.1 Chicago Accident Breakdown

A temporal analysis was used to reveal distinct patterns in crash occurrence and severity. The figure in Appendix A1 shows that weekday crashes concentrate heavily during rush hour periods (7-9 AM and 3-6 PM), while weekend crashes display a more diffused pattern throughout the day.

3.2.2 Crash and Census Overlap

In addition, a comprehensive correlation matrix (Appendix A2), was created to reveal several key relationships between demographic variables and traffic crash metrics at the zip code level. This matrix demonstrated strong positive correlations between population size and total crash count and injuries count. Additionally, a moderate positive correlation with crash rate per capita, injury rate per capita, as well as a notably strong positive correlation with the proportion of severe crashes and high damage crashes was observed from this correlation matrix.

3.2.3 Chicago Camera Breakdown

The initial analysis of Chicago's traffic camera infrastructure revealed important context for understanding enforcement patterns. The city initially began their camera implementation with the installation of their redlight cameras from 2003 to 2012. This was then followed by the deployment of speeding cameras in 2013, with an aggressive deployment of cameras in the most recent years. This historical perspective, as seen in Appendix A3, showed that camera placement and enforcement intensity vary significantly across the city and over time, which could influence both crash reporting and actual crash occurrence.

2.2.3 Redlight/Speeding Camera Analysis

The goal of this analysis was to identify the mathematical impact of red light and speeding camera implementation on crash and severe injury occurrence at the areas these cameras were implemented.

2.2.4 Data Selection

During the initial exploratory data analysis, it was discovered that the violations data available from the City of Chicago Data Portal only consisted of violation records dating back to June 2014. This was an issue because most of the city's redlight cameras were installed prior to this date, making it impossible to conduct an analysis on the impact of the cameras before and after their implementation. To ensure that the data needed to conduct a valid comparison was available, the red light and

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speeding camera analysis was restricted to only focus on the cameras installed between January 1st, 2018, and December 31st, 2024, ensuring that each camera had at least a year's worth of comprehensive violation data prior to and after their implementation.

4 METHODOLOGY

4.1 Effects of Red Light And Speed Camera Installations

To measure the safety impact, a geospatial temporal analysis was employed, where a quarter-mile buffer radius around each camera was initially created. This allowed for the isolation of the crashes that occurred within this region and which were then associated with a specific camera. The data for each camera was then segmented into two groups, based on each camera's go-live date. The "before" period consisted of all crashes occurring in a camera's buffer region prior to its go-live date, while the "after" period consisted of all crashes in a camera's buffer region after its implementation. To account for the varying lengths of these timeframes for each camera, the data was standardized by calculating the daily crash rate, which was determined by taking the total number of crashes in each period and dividing it by the number of days. The daily averages for severe injury rate were additionally obtained in a similar way.

4.2 Association of Poverty Rate with Crash Severity

Following aggregation of individual crash records and merger with 2020–2024 census-derived socio-economic indicators (population and poverty rate) at the ZIP-code level, the analysis focused on 58 Chicago ZIP codes. To ensure comparability across ZIP codes with widely differing total crash volumes, all severity and contextual factors were converted into proportions or averages. The primary outcome of interest, ***proportion_severe_crashes***, was calculated as the percentage of all crashes in each ZIP code that resulted in at least one fatal or incapacitating injury. Supporting predictor variables included ***proportion_poor_road_cond*** (percentage of crashes in which a road defect such as ruts, holes, debris on roadway, or worn surface was reported), ***proportion_inadequate_lighting*** (percentage of crashes occurring under darkness, dawn, dusk, or darkness with lights present leading to reduced-visibility conditions), ***proportion_vru_crashes*** (percentage of crashes involving at least one pedestrian or bicyclist, collectively termed vulnerable road users), and ***average_speed_limit*** (mean posted speed limit across all crashes in the ZIP code). The key socio-economic predictor, ***poverty_rate***, was the census-derived percentage of residents living below the federal poverty line. All proportion and average variables were continuous where proportion values were bounded between 0 and 1, and average speed limit variable were bounded between 0 and 100.

Statistical analysis was conducted using ordinary least squares (OLS) regression in Python's statsmodels library. OLS was selected because the outcome and all predictors were continuous, the relationships appeared approximately linear in preliminary scatterplots, and the method estimates the best-fitting linear relationship by minimizing the sum of squared residuals. A two-stage approach was adopted to balance transparency, statistical rigor, and avoidance of overfitting given the modest number of areal units (n = 57).

A **bivariate OLS model** (also called simple linear regression) examines the relationship between one continuous predictor X and one continuous outcome Y:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

where Y_i is the outcome for ZIP code i (here, proportion_severe_crashes or poverty_rate), β_0 is the intercept (expected value of Y when $X = 0$), β_1 is the slope (change in Y for a one-unit increase in X), ϵ_i is the random error term.

From each bivariate model we extracted the signed Pearson correlation coefficient $r = sign(\beta_1) \times \sqrt{R^2}$ and the two-tailed p-value for β_1 ($\alpha = 0.05$). These bivariate regressions served as an exploratory screening step and provided transparent zero-order associations.

A **multivariate OLS model** (multiple linear regression) simultaneously includes several predictors to assess their independent associations with the outcome while holding all other predictors constant:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \epsilon_i$$

Each β_j now represents the expected change in Y for a one-unit increase in predictor X_j, adjusted for all other predictors in the model. We estimated three nested multivariate OLS models with ***proportion_severe_crashes*** as the dependent variable:

Model 1:

$$proportion_severe_crashes_i = \beta_0 + \beta_1 \cdot poverty_rate_i + \epsilon_i$$

Model 2 :

$$proportion_severe_crashes_i = \beta_0 + \beta_1 \cdot poverty_rate_i + \beta_2 \cdot pro$$

Model3:

$$proportion_severe_crashes_i = \beta_0 + \beta_1 \cdot poverty_rate_i + \beta_2 \cdot pro$$

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All models used listwise deletion (n = 57). Model assumptions were assessed via R^2, adjusted R^2, Omnibus test (residual normality), Durbin–Watson statistic (residual independence), and condition number (multicollinearity).

4.3 Spatio-Temporal Hotspot Detection Using K-Means Clustering

Following the filtering of the Chicago crash dataset to include only records from January 1, 2018, onward with valid geographic coordinates, timestamps, and injury information, the analysis focused on 213,056 crashes to identify spatio-temporal hotspots. To ensure comparability across temporal segments with varying crash volumes, data was stratified by hour (0–23) and day type (weekday vs. weekend). The primary outcome of interest, crash hotspots, was defined as spatially dense clusters enriched with severity metrics. Supporting variables included latitude and longitude for spatial grouping***, crash_hour*** for temporal slicing, a binary ***is_weekend*** flag (1 if ***day_of_week*** is Saturday or Sunday), injuries_total (sum of all reported injuries), and ***injuries_severe*** (sum of fatal and incapacitating injuries). All variables were derived from the raw dataset, with boolean flags for ***any_injury*** and ***severe_injury*** to facilitate severity overlays. The key clustering inputs, geospatial coordinates, were continuous and bounded by Chicago's extents.

Unsupervised clustering was conducted using **K-Means** in Python's scikit-learn library. K-Means was selected because it efficiently partitions data into k clusters by minimizing within-cluster variance, handles large geospatial datasets, produces interpretable centroids as hotspot centers, ensures consistent cluster counts across frames (unlike density-based methods like DBSCAN), and allows flexible parameterization for animation. A multi-stage approach was adopted to balance computational efficiency, interpretability, and granularity given the dataset size (n ≈ 213,056.

A baseline K-Means model (independent per hour and scenario) assigns each point to the nearest centroid:

$$argmin_S \sum_{i=1}^{k} \sum_{x \in S_i} \|x - \mu_i\|^2$$

where $S = \{S_1, ..., S_k\}$is the set of clusters, $x$is a data point (latitude, longitude), and $\mu_i$is the centroid of cluster S_i. For each hour h and scenario (total injuries vs. severe injuries), k was drawn from {3, 5, 7, 10, 20, 30}, with adaptive adjustment $k' = min(k, n_h)$where $n_h$is the number of points in hour h. These independent runs served as an exploratory step and provided visualizations at varying resolutions.

A validation step using Silhouette Score assessed cluster quality:

$$s(x) = \frac{b(x) - a(x)}{max(a(x), b(x))}$$

where $a(x)$is the mean distance to points in the same cluster, and $b(x)$is the mean distance to the nearest other cluster. As shown in Appendix O, scores were highest for k=4–7 (strong separation) and lower for k>20 (finer granularity for neighborhood analysis). Model assumptions were verified via elbow plots for inertia and visual inspection of cluster stability.

Post-clustering, centroids were enriched with:
(1) Crash Count per Cluster

$$CrashCount_i = |C_i|$$

(2) Injury Severity Metric
For each cluster :

$$Severity_i = \sum_{x \in C_i} InjuryWeight(x)$$

Where InjuryWeight is:

- $INJURIES_TOTAL$ for the first map
- $INJURIES_SEVERE$ for severe-hotspot overlay.

These metrics were normalized for visualization:

$$ColorIntensity_i = \frac{Severity_i}{max(Severity\ across\ clusters)}$$

Thus, larger markers = more crashes, darker colors = more severe injuries

Visualizations were constructed using Plotly for interactive geospatial heatmaps. For each hour h=0 to 23, frames included centroids with marker size ∝ CrashCount (density), color ∝ ColorIntensity (severity), and hover data (crashes, injuries, type). Frames were animated with sliders and play controls; dropdowns toggled k and layers (total/severe, weekday/weekend). This approach captures temporal dynamics (e.g., rush-hour volume vs. late-night severity) and day-type shifts, supporting scalable, reproducible insights for safety planning.

4.4 Predictive Model for Crash Severity

The primary goal of the modeling in this study is to predict whether a reported crash is severe (involving injury or fatality) or non-severe, using information available shortly after the crash is recorded. The target variable is highly imbalanced, with only about 1.66% of records labeled as severe (1) and 98.34% as non-severe (0). To address this, the modeling pipeline was structured in three stages: baseline models for assessing problem difficulty, advanced models for improved performance, and decision-threshold tuning for optimization.

Initial baseline models included a 'DummyClassifier', which predicts the majority class (non-severe) yielding ~98.4% accuracy but zero precision and recall for the severe class, resulting in a PR-AUC near the positive rate (~0.016); this highlighted the inadequacy of accuracy as a metric and the need for PR-AUC, recall, and F-scores. Logistic Regression with one-hot encoding for categorical variables and balanced class weights achieved a PR-AUC of ~0.146, indicating meaningful

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linear patterns. A Random Forest baseline, also with one-hot encoding and balanced weights, captured nonlinear interactions and reached a PR-AUC of ~0.134, comparable to Logistic Regression but with higher computational demands. These baselines confirmed the problem's learnability and justified advancing to gradient boosting trees, which excel on large tabular datasets.

For the final model, LightGBM was selected as a gradient boosting framework that constructs shallow decision trees sequentially, with each tree correcting prior errors through leaf-wise growth to efficiently handle complex interactions. Key selection reasons include its strong performance on large tabular data with mixed numerical and categorical features, automatic management of nonlinear relationships and interactions, built-in support for categorical features and class weights, and fast training suitable for the ~950,000-row dataset. The default model used binary objective, standard tree and leaf counts, early stopping after 100 rounds based on validation AUC to prevent overfitting (stopping if validation AUC fails to improve for 100 iterations and retaining the best iteration), and categorical columns cast to category dtype for proper handling. A tuned version incorporated parameters like n_estimators=1000, learning_rate=0.05, num_leaves=127, feature_fraction=0.7, bagging_fraction=0.8, and bagging_freq=5; however, it showed no meaningful performance gains over defaults, as LightGBM's standard settings are robust for this data type, with imbalance posing the primary limitation rather than model capacity.

Class imbalance was addressed through two strategies. First, class weights were tested with class_weight=None (no weighting) and class_weight="balanced," which slightly altered raw prediction scores but yielded similar PR-AUC and precision-recall trade-offs post-threshold tuning, offering no substantial improvement. Second, resampling involved oversampling the severe class to ratios of 10–30% and training both weighted and unweighted models on resampled data; however, these did not outperform the original unresampled dataset and occasionally added noise. Consequently, the final model relies on the original dataset without resampling.

After training, the top 20 most important features were analyzed using LightGBM's built-in "gain" metric, which quantifies each feature's contribution to error reduction across trees. Rankings were generated for both unweighted and balanced-weight models, revealing consistent influential features in both, demonstrating model stability and minimal impact from class weighting on core learning behavior. This interpretation step confirms the model learns reasonable relationships, ensuring it is not only accurate but also explainable.

5. RESULTS AND DISCUSSION

5.1 Redlight and Speeding Camera Analysis Results

The average percent change in daily crashes as well as severe injuries for each camera were plotted in a box-and-whisker plot (Appendix B1) to determine the overall impact of camera implementations. From this graph, there was an increase in both accident rate, as well as injury severity, after a camera had been implemented in a certain area. After we had obtained this data, a Wilcoxson Signed-Rank Test was utilized to determine whether this was a statistically significant difference in crash and severe injury rates before and after the camera's go-live. The decision to use a Wilcoxson Signed-Rank test was due the crash frequency data not following a normal distribution, therefore making a standard paired t-test inappropriate for this situation.

$$T^+ = \sum_{D_i > 0} R_i \quad \text{and} \quad T^- = \sum_{D_i < 0} R_i \tag{1}$$

$$W = \min(T^+, T^-) \tag{2}$$

For large sample sizes (N > 20), the distribution of W becomes approximately Normal (bell-shaped). From this, a Z-score can be calculated from this value and used to find the probability that the data rejects the null hypothesis that this increase in crash and severe injuries was due to random chance. While looking at Severe injuries, this test produced a p-value of 0.8402, which fails to reject the null hypothesis. This suggests that for the cameras analyzed, there was no statistically significant difference in the rate of severe injuries before and after a camera had been implemented.

On the other hand, while observing the crash frequency, statistical testing resulted in a p-value of 0, which rejects our null hypothesis. For every pair of camera data points, each camera resulted in an increase in daily average crashes after their implementation. Because this p-value was less than the 0.05 required to reject the null hypothesis, the results suggest that this increase in crash rate after a camera's implementation is most-likely not due to random chance.

5.2 Association of Poverty Rate with Crash Severity Results

Bivariate analyses revealed stark differences in crude associations with proportion of severe crashes as shown in Table 1.

Predictor	Pearson r	p-value	Interpretation
Poverty rate	+0.724	0.00	Very strong positive

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Proportion poor road condition	+0.361	0.01	Moderate positive
Proportion inadequate lighting	+0.396	0.00	Moderate positive
Proportion VRU crashes	–0.406	0.00	Strong protective
Average speed limit	–0.012	0.93	No association

TABLE 1:BIVARIATE ASSOCIATIONS WITH PROPORTION OF SEVERE CRASHES

Poverty rate exhibited a very strong positive association ($r = +0.724$, $p < 0.001$). Poor road condition and inadequate lighting showed moderate positive associations ($r = +0.361$ and $+0.396$, respectively; both $p \leq 0.005$). A higher proportion of crashes involving vulnerable road users (pedestrians and bicyclists) displayed a strong protective association ($r = –0.406$, $p < 0.001$), whereas average posted speed limit had essentially no linear association ($r = –0.021$, $p = 0.93$).

Predictor	**Model 1 β (SE)**	**Model 2 β (SE)**	**Model 3β(SE)**	**Model 3 p-values**
Poverty rate	0.0361	0.0321	0.0315	0.000
Proportion poor road condition	–	0.0418	0.0231	0.682
Proportion inadequate lighting	–	0.0149	0.0128	0.218
Proportion VRU crashes	–	–	-0.0486	0.032
Average speed limit	–	–	0.0009	0.075
Adjusted R^2	0.516	0.524	0.566	

TABLE 2:NESTED ORDINARY LEAST SQUARES REGRESSION MODELS PREDICTING PROPORTION OF SEVERE CRASHES (N = 57)

Poverty rate remained highly significant and remarkably stable across all specifications. In the fully adjusted model (Model 3), a one percentage-point increase in the poverty rate was independently associated with a 0.0315 percentage-point increase in the proportion of severe crashes (95% CI: 0.021–0.042). Over the observed range of poverty rates in Chicago (approximately 5% to 50%), this translates into an estimated difference of more than 1.4 percentage points in severe-crash proportion between the wealthiest and poorest neighborhoods which is a substantial disparity given the city-wide average of around 1.5–2%.

The apparent effects of poor road conditions and inadequate lighting observed in bivariate analyses were eliminated once poverty_rate was controlled ($p = 0.682$ and $p = 0.218$, respectively), indicating complete mediation. In contrast, the proportion of crashes involving vulnerable road users retained a significant independent protective effect ($\beta = –0.0486$, $p = 0.032$). A 10 percentage-point higher share of pedestrian and bicyclist crashes was associated with approximately half a percentage-point lower severe-crash proportion, even after accounting for poverty and infrastructure quality. Average posted speed limit exhibited no meaningful independent association with severity at the ZIP-code level ($p = 0.075$).

5.3 Spatio-Temporal Hotspot Detection Results

Hourly aggregation revealed strong temporal variation in both crash frequency and severity. The highest severe crash counts occurred during the afternoon and early evening hours (15:00–19:00), corresponding to periods of elevated traffic volume and congestion. However, a divergence was observed when examining the severity rate. The highest severe-injury rates occurred during late-night and early-morning hours (00:00–04:00), peaking at 02:00 (2.89%). This suggests that while fewer crashes occur late at night, they are significantly more likely to be severe, likely due to higher operating speeds and reduced visibility.

The hourly hotspot maps, referenced in Appendix P, reflected this dichotomy. Midday and afternoon frames showed large marker sizes and intense coloration, reflecting the high absolute number of injuries during peak travel times. In contrast, late-night frames displayed smaller and lighter clusters due to reduced traffic volume, though the proportion of severe outcomes remains a critical safety concern.

The study detected meaningful behavioral differences between weekdays and weekends. Weekdays accounted for 155,451 crashes, of which 1.29% were severe. Weekends, while representing fewer total crashes, showed a noticeably higher severe-crash proportion of 1.74%. This indicates that weekend crashes are approximately 35% more likely to be severe compared to weekday crashes. Spatially, the difference is distinct. As visualized in Appendix Q, weekday hotspots tended to concentrate along major commuting corridors. In contrast, weekend hotspots shifted toward entertainment districts and recreational centers. The weekend visualization highlights specific clusters with elevated severity intensity, underscoring location-specific risks that differ from the Monday–Friday commute patterns.

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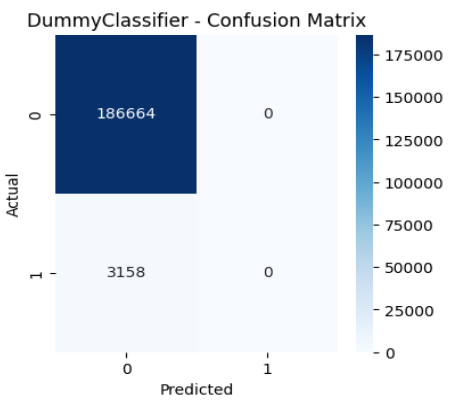
The interactive hotspot maps generated through the K-Means clustering framework provided a clear, interpretable representation of complex spatio-temporal crash patterns. The visual distinction, where size equals frequency and color equals severity, allows for the immediate identification of "high-risk" versus "high-volume" intersections. The validation of these clusters via Silhouette scores (Appendix O) confirms that the spatial groupings are distinct, providing confidence in the detected hotspots. These findings support practical applications such as optimizing police patrol routes, EMS positioning, and Vision Zero safety projects.

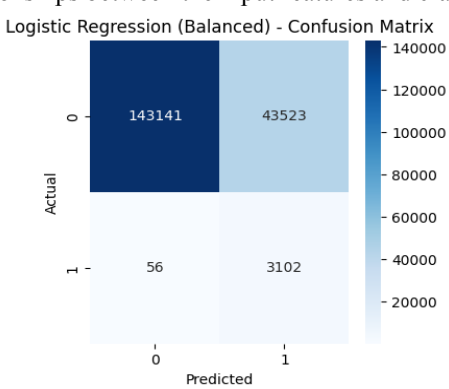
5.4 Predictive Model for Crash Severity Results

When compared to several baseline models to understand how hard it is to predict severe crashes , the DummyClassifier always predicts the majority class, achieved very high accuracy (~98.4%) but completely failed to identify any severe crashes. Its PR-AUC was close to the severe crash rate (~0.016), which confirms that accuracy alone is not a good metric for this imbalanced problem.



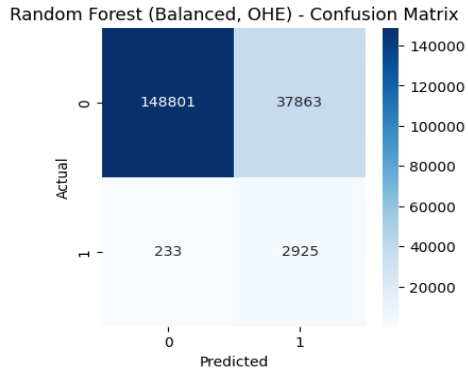
Appendix F : Baseline Dummy Classifier Confusion Matrix

Logistic Regression with one-hot encoding and balanced class weights significantly improved performance, reaching a PR-AUC of about **0.146**. This shows that there are clear linear relationships between the input features and crash severity.



Appendix G : Logistic Regression Model Confusion Matrix

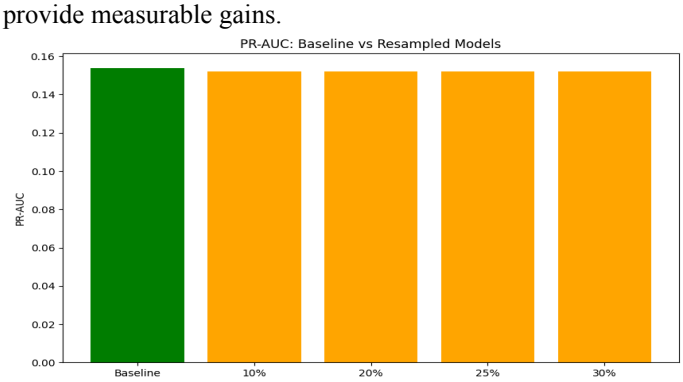
Random Forest, which can capture nonlinear patterns, produced a PR-AUC of around **0.134**. Although it models interactions between variables, it did not outperform Logistic Regression by a large margin and required more computation.



Appendix H : Random Forest Model Confusion Matrix

LightGBM gave the best overall performance, with PR-AUC around **0.153** on the test set. This confirms that gradient boosting trees are well suited for this kind of large, mixed-type traffic data. When we tried various hyperparameter settings (changing learning rate, number of leaves, and minimum child samples), the PR-AUC stayed in a narrow range around the baseline value. The tuning trials did not produce a clear improvement over the default configuration, which suggests that the defaults are already close to optimal and that the main challenge is the inherent noise and overlap between severe and non-severe crashes.

We also observed that class weighting and resampling did not provide measurable gains.



Appendix I: Tuned LightGBM Model Performance Comparison Plot

Both class_weight="balanced" and different oversampling ratios produced models with PR-AUC and F_2 scores very similar to the original unweighted model.

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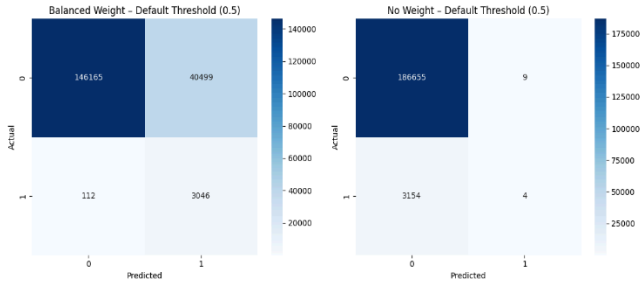
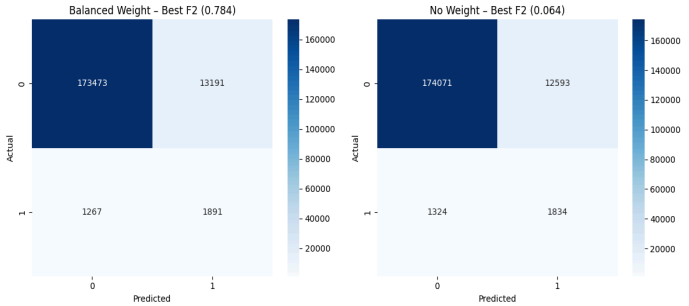
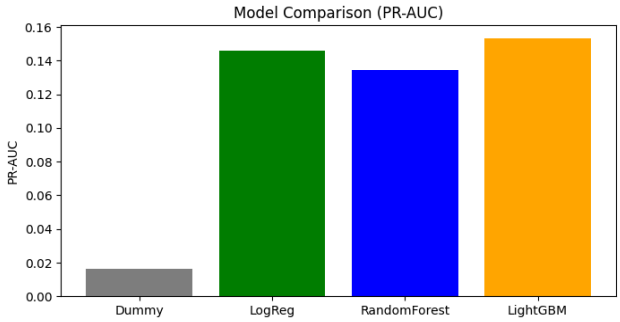


Appendix J: Confusion Matrix at Default Threshold (0.50) (Balanced Class Weight vs No Class Weight)



Appendix K: Confusion Matrix at BestF2 Score (Balanced Class Weight vs No Class Weight)

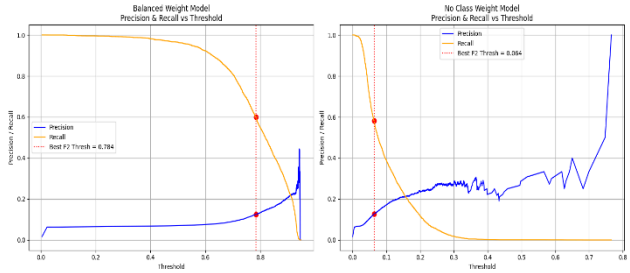
After threshold tuning, the precision–recall curves of the weighted and unweighted models almost overlapped. This indicates that LightGBM is already learning good probability estimates on the original data distribution, and further manipulation of class frequencies does not help much.



Appendix L : PR-Curve Comparison between Models

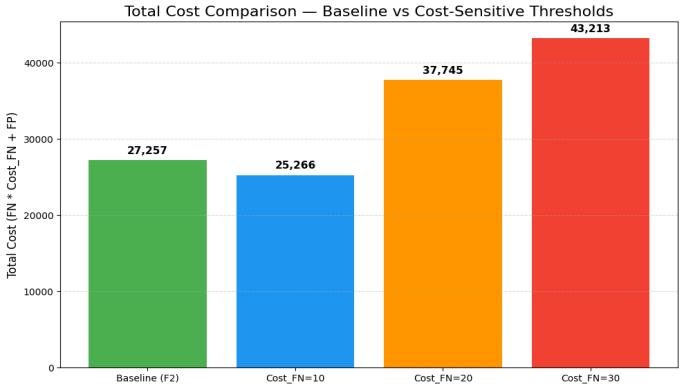
The most important improvement came from **threshold selection** rather than from changing the model itself. At the default 0.50 threshold, the LightGBM model achieved extremely high specificity for non-severe crashes, but recall for severe crashes was close to zero, which is unacceptable in a safety context.

After applying F_2-based threshold tuning and choosing a threshold of 0.061, recall for severe crashes increased to above **55%**, while precision for severe cases stayed around **12–13%**. This means the model can capture more than half of the severe crashes, with a manageable number of false alarms. From a practical perspective, this is a much more useful operating point: traffic analysts or emergency services can review a moderate list of high-risk crashes instead of missing most of them.



Appendix M: Threshold Tuning Curve (balanced class weight vs no class weight (F2-Score vs Threshold)

The cost-sensitive analysis supports this conclusion. As we increased the cost of missing a severe crash, the optimal threshold moved lower (0.079 → 0.044 → 0.030), which reduced false negatives but increased false positives.



Appendix N: Base Model Comparing with Cost-Sensitive Thresholds

These cost-based thresholds were in the same range as the F_2-optimal threshold, which gives confidence that our chosen threshold is reasonable from both a statistical and an operational point of view.

Top 20 Feature Importance

The top 20 feature importance results show that LightGBM is consistently focusing on meaningful variables that align with real-world crash severity patterns. Across both the unweighted and class-weighted models, the same features appear at the top, although the exact order varies slightly. This consistency indicates that class weighting has very little effect on the model's internal representation, and both models learn almost identical relationships from the data.

The model places strong emphasis on spatial patterns, temporal characteristics, crash mechanism details, and environmental conditions, which are all realistic drivers of severity in traffic incidents. The similarity between the two important charts also supports the earlier finding that class weighting does not materially improve performance. Overall, the feature importance analysis confirms that LightGBM is learning

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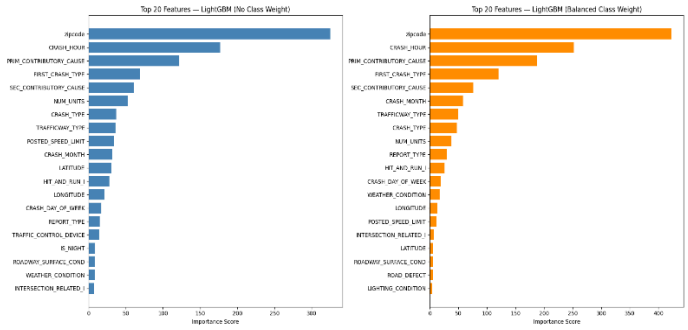
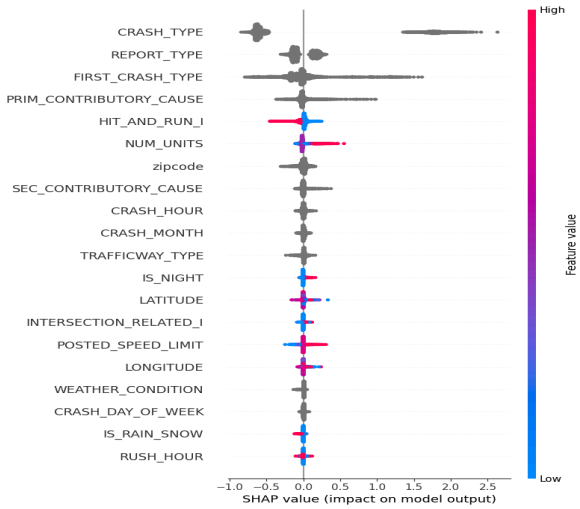
sensible and interpretable patterns, giving confidence that the final model is not only performing well but is also aligned with domain knowledge in road-safety analytics.



Appendix O: Feature Importance Plot – Weighted/Unweighted Model

The SHAP summary plot provides a deeper explanation of how individual features influence the LightGBM model's prediction of crash severity. Unlike feature importance, which only shows which variables are used the most, SHAP shows how each feature pushes the prediction higher or lower for every crash record.



Appendix P: SHAP Summary Plot for LightGBM

In the plot, points on the right side (positive SHAP values) indicate a higher chance of the crash being severe, and points on the left side indicate a lower chance. We observe that features related to crash mechanisms such as CRASH_TYPE, FIRST_CRASH_TYPE, and PRIM_CONTRIBUTORY_CAUSE, have the strongest and most consistent impact on severity. These variables show wide horizontal spread, meaning they can push predictions strongly in both directions depending on the category. Features like NUM_UNITS, HIT_AND_RUN_I, CRASH_HOUR, and SEC_CONTRIBUTORY_CAUSE also play meaningful roles, indicating that time of crash and the nature of violation influence severity. Environmental features such as WEATHER_CONDITION,IS_NIGHT,POSTED_SPEED_LIMIT, and ROAD SURFACE show smaller but still visible contributions. Overall, the SHAP analysis confirms that the model is learning logical patterns, the way the crash occurred, what caused it, and when it happened matter more for severity prediction than simple geographical or demographic factors. This interpretability step helps to validate that the model's behavior matches real-world understanding of traffic safety.

Overall, the results demonstrate that severe crash predictions are feasible but challenging due to strong class imbalance and overlapping feature distributions. LightGBM proves to be a suitable model for this task, clearly outperforming naive baselines. Hyperparameter tuning, class weighting, and resampling yield only minor changes in metrics. Decision-threshold tuning using the F_2 score emerges as the key step that transforms the model into a useful tool for prioritizing severe crashes.

6. RECOMMENDATIONS AND CONCLUSION

Place 3-4 line conclusion here.

7. ACKNOWLEDGEMENTS

The authors would like to express their sincere gratitude to the City of Chicago for maintaining and providing open access to comprehensive traffic crash data through the Chicago Data Portal. We are equally grateful to the U.S. Census Bureau for making demographic and socioeconomic data publicly available through the American Community Survey, which enabled the zip code analysis components of this study.

We extend our appreciation to our capstone professor for his guidance and valuable feedback throughout the development of this project. His expertise and support were valuable in shaping the direction of this research.

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Finally, we acknowledge the broader open data movement and the many agencies and organizations that recognize the value of transparent, accessible public data in advancing research for the public good.

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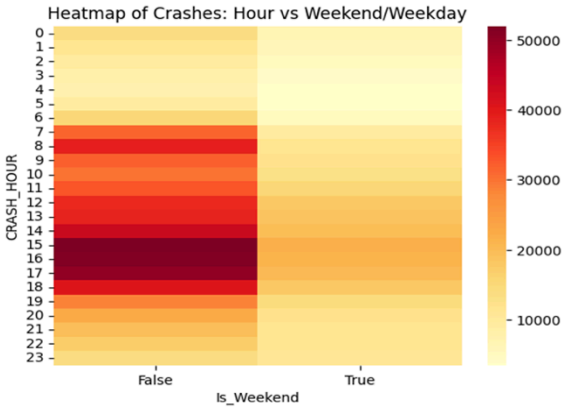
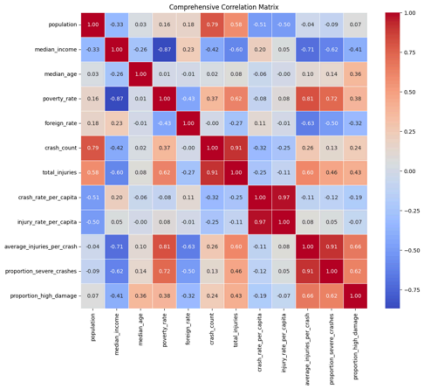
APPENDIX

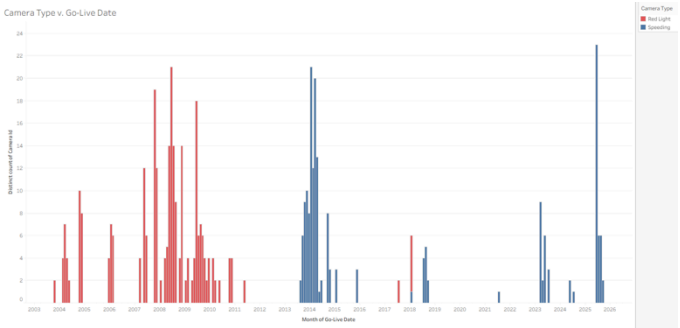
A1. FIGURE: HEATMAP OF CRASH OCCURRENCE VS WEEKEND BINARY



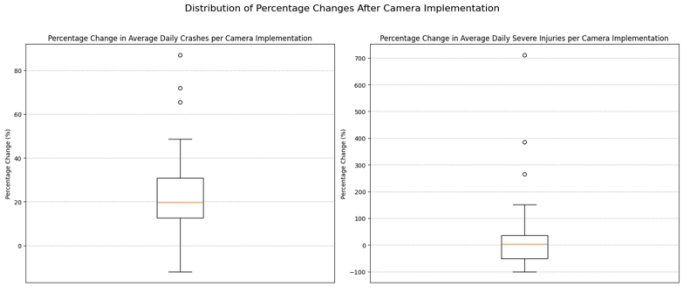
A2. FIGURE: CORRELATION MATRIX BETWEEN CENSUS VARIABLE AND CRASH/INJURY RATE



A3. FIGURE: BAR GRAPH DEPICTING CAMERA TYPE VS IMPLEMENTATION DATE



B1. FIGURE: BOX-WHISKER PLOT DEPICTING PERCENT CHANGES OF DAILY CRASHES AND SEVERE INJURIES AFTER CAMERA IMPLEMENTATION



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